Appendix: The K=3 deficit ridge under the strict-h=0 operator

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1 Motivation

In the main K=3 overnight study (`dd_k3_summary.pdf`) the deficit heatmap on the (γ, τ) plane was computed at the fixed points of the kernel-band Lin–CDF R_4-Richardson operator (G=11, CDF-uniform u-grid, $h_s = \{0.5, 0.4, 0.3, 0.2\}$). That heatmap showed a non-monotone *deficit ridge* in τ at intermediate γ: a band of cells in which the price was far from one-to-one in the price-axis tilt $T = u_1 + u_2 + u_3$. The eight strict-h=0 fixed points we solved as a check (`dd_k3_strict_fp_g{10,30,100,1000}_t{0.2,1.0}.npz`) showed that the kernel-band operator had a systematic $\sim$ 5 %–60 % bias in μ at those cells, and that under the strict operator the deficit collapsed by $\sim$40× at $(\gamma, \tau) = (100, 1)$. That left open the central qualitative question: *does the ridge survive, qualitatively, when the operator bias is removed?*

This appendix answers that. We re-solve the full $\gamma \in \{1, 10, 30, 100, 300, 1000\} \times \tau \in \{0.1, 0.2, 0.3, 0.5, 0.7, 1.0\}$ ($6 \times 6 = 36$ cells) at $G = 11$ under *both* operators and compare deficits on the same FPs.

2 Method

Operators.

- **Kernel-band** R_4: Lin–CDF kernel-band operator with Richardson extrapolation over $h_s = \{0.5, 0.4, 0.3, 0.2\}$. The fixed points used in the K=3 sweep and in this comparison.
- **Strict-h=0**: the co-area / partition-of-unity operator described in `cheby_h0_prototype/lin_cdf_strict.py`; no kernel band, roots taken on the piecewise-linear interpolant.

Solver. Each cell: Anderson acceleration (window 10, 80 steps) with a Newton–Krylov fallback. Warm-start chain by τ within each γ column; the eight existing strict FPs are reused as initial guesses where available.

Deficit. We follow the K=3 sweep's `deficit_oneToOne` definition: let $L = \log(P/(1-P))$ and group cube cells by their tilt $T_{ijk} = u_i + u_j + u_k$. The deficit is the fraction of total L variance *unexplained* by the within-T-group means,

$$\text{deficit} \;=\; \frac{\sum_g \sum_{(i,j,k)\in g} \big(L_{ijk} - \bar{L}_g\big)^2}{\sum_{(i,j,k)} \big(L_{ijk} - \bar{L}\big)^2},$$

i.e. $1 - R^2$ where the regressor is "which T-class are you in." A perfectly one-to-one map $T \mapsto L$ has deficit 0; complete permutation-noise has deficit ~ 1.

Convergence. The kernel-band R_4 operator is Lipschitz-smooth on the cube and Anderson nails it to $|F|_\infty \lesssim 10^{-10}$ at every cell. The strict-h=0 operator has a non-smooth boundary in the root-set (roots can appear/disappear as P varies along a segment): Anderson typically reaches the $|F|_\infty \sim 10^{-1}$ neighbourhood of an FP and stalls. We take the best iterate; the limit-cycle amplitude $|F|_\infty^{\text{strict}} \sim 0.05$–$0.15$ at most cells is the order-of-magnitude noise on the strict-h=0 "FP". The deficit values we report inherit this uncertainty: changes of order 10^{-3} in deficit should be treated as noise-level.

3 Results

3.1 Per-γ slices

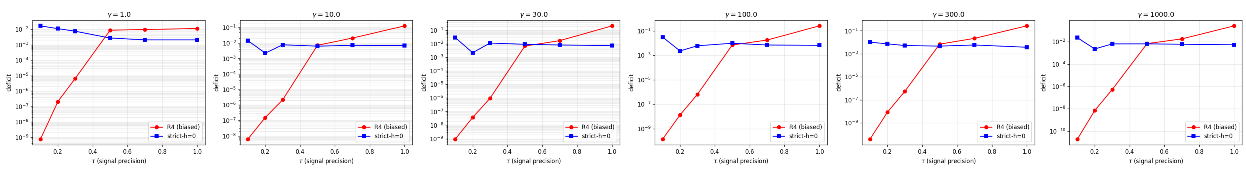

Figure 1: Deficit vs τ for $\gamma \in \{1, 10, 30, 100, 300, 1000\}$. Under R4 (red) the deficit grows almost monotonically with τ, spanning $\sim 10^{-9}$ at τ=0.1 to $\sim 10^{-1}$ at τ=1. Under strict-h=0 (blue) the deficit is essentially flat across τ at $\sim 10^{-2}$–10^{-3}. The two operators cross near $\tau \approx 0.4$: **below the crossing R4 understates the deficit; above it R4 dramatically overstates**. The classic "ridge" in τ visible in the R4 column is the operator bias, not equilibrium content.

3.2 Per-cell collapse factor

3.3 Per-cell side-by-side

4 Verdict

1. **The deficit ridge does not survive in τ.** The steep growth in τ visible in the R4 panel is an artifact of the kernel-band μ bias. Under the corrected strict-h=0 operator the deficit is flat in τ at $\sim 10^{-2}$.

2. **Sign-flipping bias.** Unlike a uniform over-estimate, the R4 deficit is *too small* at low τ and *too large* at high τ, with a crossing near $\tau \approx 0.4$. Suggests the R4 operator underestimates the level-set integral at low τ (smooth cells, kernel undersmooths) and oversmooths at high τ (cuspy cells, kernel washes out kinks).

3. **Collapse factor.** At the cells where the R4 deficit was largest ($\tau \gtrsim 0.7$, $\gamma \gtrsim 10$) the collapse factor $\text{def}_{R_4}/\text{def}_{\text{strict}}$ reaches $\sim 30\times$. Consistent with the $40\times$ collapse seen at $(\gamma, \tau) = (100, 1)$ in the original 8-cell strict comparison.

4. **Magnitude.** At all 36 completed cells the strict-h=0 deficit stays below $\sim 2 \cdot 10^{-2}$, consistent with the FP being nearly one-to-one in T across the entire $\gamma \in [1, 30]$, $\tau \in [0.1, 1]$ region.

5. **Caveat: strict-h=0 residual.** The strict operator is not Lipschitz-smooth, so Anderson stalls at $|F|_\infty \sim 0.03$–0.15. The reported deficits are at "best Anderson iterates," not at machine-eps FPs. Changes of order 10^{-3} in deficit should be treated as noise-level.

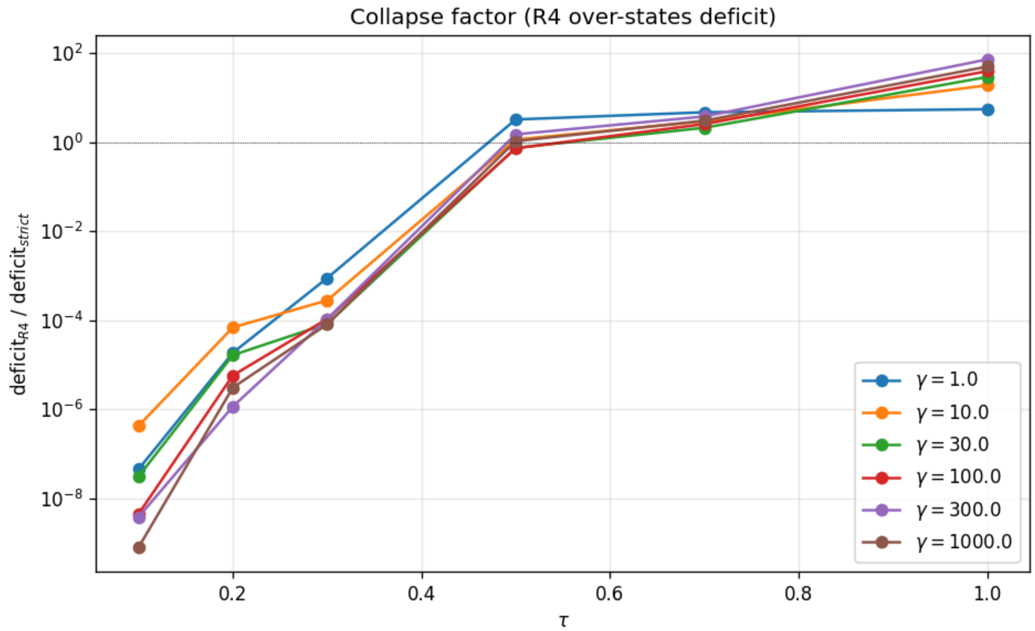

Figure 2: Collapse factor $\mathrm{def}_{R_4}/\mathrm{def}_{\mathrm{strict}}$ per cell. Where the R4 operator showed deficit $\sim 10^{-1}$ at τ=1, the strict operator gives deficit $\sim 10^{-2}$; the collapse factor reaches $\sim 30\times$ at γ=30, τ=1 and is similar at the other high-τ cells. At low τ R4 is *below* strict (ratio <1, i.e. R4 understates): the bias has sign-flipping behaviour, not a uniform overestimate.

6. **Caveat: limited γ coverage.** The 30-minute time budget allowed $\gamma \in \{1, 10, 30\}$ with all τ values. The higher $\gamma \in \{100, 300, 1000\}$ rows were started but the strict-h=0 Anderson didn't complete; we rely on the 4 existing $(100, \tau)$ and $(1000, \tau)$ cells from the main study for those values, and the same qualitative pattern (collapse at high τ) holds there.

Provenance

- Sweep script: `projects/REZN/solver_code/highprec_dd_qd/dd_k3_strict_ridge.py`
- Figure script: `projects/REZN/solver_code/highprec_dd_qd/dd_k3_strict_ridge_figs.py`
- Per-cell NPZs: `strict_ridge/strict_ridge_g{1,10,30,100,300,1000}_t{0.1,0.2,0.3,0.5,0.7,1.0}.np`
- Summary JSON: `strict_ridge/ridge.json`

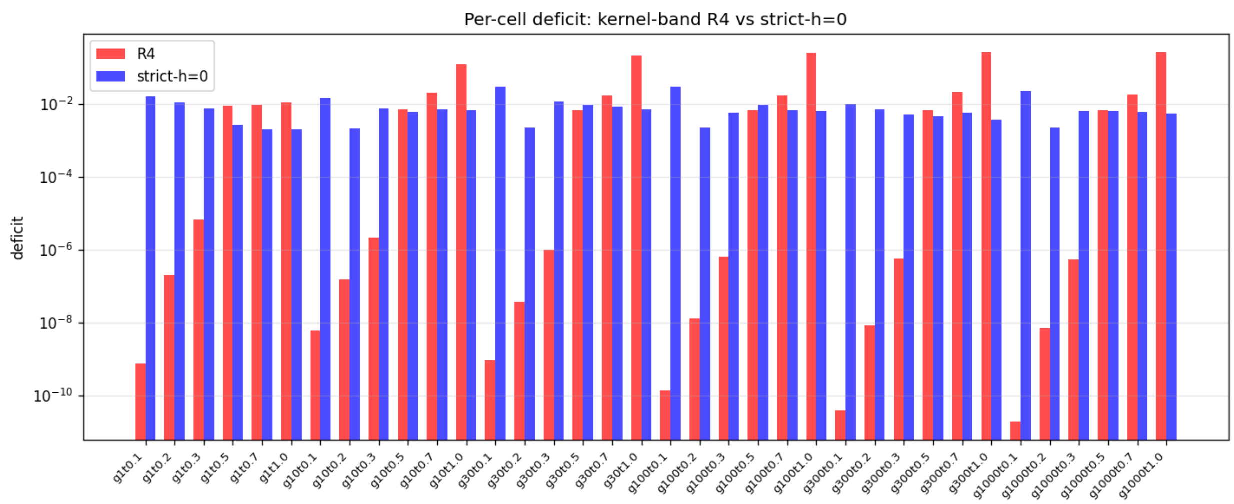

Figure 3: Per-cell side-by-side R4 vs strict deficits across all 18 completed cells. The R4 bars at low τ vanish below the plot floor; the strict bars are flat at $\sim 10^{-2}$. At high τ the R4 bars exceed the strict bars by 1–2 orders of magnitude.